2017-Winter Paper 3 – Subspeciality of Urology

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Q1. Describe embryology of penis. Describe management of proximal hypospadias.

Type: LAQ (25 marks)

A. Embryology of the penis

Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

Hormonal control

- SRY (sex-determining region Y) → testis → **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

Primordium (germ layer)	Male	Female
Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce)	Glans, corpora	Clitoris
Urethral folds	Fuse → penile urethra, median raphe	Labia minora
Labioscrotal swellings	Scrotum	Labia majora
Urethral plate (endoderm)	Whole urethra, including glanular	Vestibule

Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe.

- The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated.

- Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks.
- **Prepuce** fuses dorsally first; ventral fusion awaits urethral closure, hence the **dorsal hood** in hypospadias.
- Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site).
- **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

B. Proximal hypospadias: evaluation

- Meatus proximal shaft, penoscrotal, scrotal or perineal (10–25%). Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone). A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- No routine VCUG (voiding cystourethrogram) or renal ultrasound; prostatic utricle commoner.

C. Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects transient; evidence mixed, but two RCTs (randomised controlled trials) showed fewer complications.

D. Curvature correction (decides one- vs two-stage)

```
Deglove + excise dysplastic dartos + artificial erection
   ├─ straight        → urethroplasty
   ├─ residual <30°   → dorsal midline plication (12 o'clock)
   └─ >30°            → divide urethral plate
                            └─ still >30° → ventral corporotomy ± graft → staged repair
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	Dorsal plication (Nesbit, Baskin)	Ventral lengthening
Method	Shortens the convex dorsal side	1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft
Use	Mild (≤15–30°)	Moderate–severe
Recurrence	28%	9%

E. Urethroplasty

Approach	Technique	Comment
One-stage **proximal TIP (tubularised incised plate)**	Plate kept, incised and tubularised	Recurrent curvature, stricture: largely abandoned for severe cases

Approach	Technique	Comment
One-stage **onlay island flap**	Inner preputial flap on preserved plate	Diverticulum, fistula
One-stage **TPIF (transverse preputial island flap, Duckett tube)**	Tube after plate division	High complications
Two-stage Bracka / STAG (staged tubularised autograft)	Stage 1: correct curvature, divide plate, quilt inner-prepuce or buccal graft. Stage 2 (≥6 months): tubularise, barrier layer	**Preferred for most proximal cases**; STAG lowest complications (about 23%)
Two-stage Byars	Dorsal preputial flaps moved ventrally, tubularised later	Complications 43–81%

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; scrotoplasty for transposition.

F. Complications and outcomes

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** commonest (repair ≥6 months later; exclude distal obstruction), glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, diverticulum (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans).
- A long, rigid neourethra raises resistance (Poiseuille), causing fistula and diverticulum.
- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

Diagram to draw: double-zipper urethral formation; curvature algorithm.

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias plus undescended testis → DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka preferred; fistula commonest complication.

Q2. Describe anatomy of pelvic floor in females and describe surgeries for SUI in females.

Type: LAQ (25 marks)

Introduction

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, coughing or sneezing. Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.

A. Anatomy of the female pelvic floor

1. Pelvic diaphragm

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal angle
Pubococcygeus (pubovaginal)	Pubis, ATFP (arcus tendineus fasciae pelvis) → viscera, raphe	Slings urethra and vagina
Iliococcygeus	Arcus tendineus levator ani, ischial spine → coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine → sacrum, coccyx	Lies on sacrospinous ligament

- Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, fascia then carries the load.

2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

4. DeLancey levels (1992)

Level	Structures	Defect
I suspension	Cardinal–uterosacral	Apical prolapse, enterocele
II attachment	ATFP, levator fascia	Cystocele (central or paravaginal), rectocele
III fusion	Perineal membrane, perineal body	**Urethral hypermobility**, perineal descent

5. Continence mechanism

- Continence zone: proximal and mid-urethra. **Oestrogen-dependent vascular submucosa** gives much of closure pressure; smooth and striated sphincters; pudendal nerve (S2–4) drives the guarding reflex.

Theory	Mechanism	Surgical answer
Pressure transmission (Enhorning 1961)	Hypermobile urethra leaves the abdominal pressure zone	Burch
Hammock (DeLancey 1994)	Urethra compressed against a firm suburethral layer	Restore backing

Theory	Mechanism	Surgical answer
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at mid-urethra	Mid-urethral sling
ISD (intrinsic sphincter deficiency)	Urethra fails to close (VLPP (Valsalva leak point pressure) <60, MUCP (maximum urethral closure pressure) <20 cmH_2O)	Sling, bulking, AUS (artificial urinary sphincter)

B. Before surgery

- Index patient: cough stress test, PVR (post-void residual), urinalysis; urodynamics if SUI not demonstrated or complicated. **PFMT (pelvic floor muscle training)**, weight loss, pessary first.

C. Surgeries for SUI

Operation	Principle	Key facts
MUS (mid-urethral sling): retropubic TVT (tension-free vaginal tape, Ulmsten 1995); transobturator (Delorme 2001; inside-out de Leval)	Loose type I polypropylene tape under the **mid-urethra**; urethra kinks on it during stress	**First line** (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) strong recommendation). **TOMUS**: 80.8% vs 77.7%, equal
SIMS (single-incision mini-sling)	Vaginal only	Non-inferior to transobturator for subjective cure; conditional recommendation (2023)
Autologous PVS (pubovaginal sling)	Rectus fascia or fascia lata at the **bladder neck**, minimal tension	All types including **ISD**, failed MUS, mesh-averse. **SISTEr**: sling beat Burch (47% vs 38%) with more voiding dysfunction and UTI (urinary tract infection)
Burch colposuspension (open, laparoscopic, robotic)	Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's ligament**	85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD
Paravaginal repair, VOS (vagino-obturator shelf)	Vagina to ATFP or obturator fascia	Flatter, less obstructive
MMK (Marshall–Marchetti–Krantz)	Bladder neck to symphysis periosteum	Osteitis pubis 0.7–2.5%; abandoned
Bulking agents	Submucosal injection improves coaptation	Frail, minimal surgery; repeat injections
AUS	Bladder-neck cuff	Second line for severe recurrent ISD; erosion, revision

- Needle suspensions abandoned (pull-through); anterior colporrhaphy is not an SUI operation.

Burch steps: Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

Retropubic vs transobturator MUS

	Retropubic	Transobturator
Bladder perforation	4.5%	0.6%
Voiding dysfunction	More (TOMUS surgery 2.7% vs 0%)	Less
Groin pain	1.3%	6.4%

	Retropubic	Transobturator
ISD, long-term durability	Often preferred	Possibly more repeat surgery

D. Complications

- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal) → urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS (International Urogynecological Association/International Continence Society) terms): exposure, extrusion, perforation into bladder or urethra, pain → excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.
- **Also know:** CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension).

Diagram to draw: DeLancey levels and hammock cross-section with Burch, bladder-neck and mid-urethral sling sites.

> **Key points to remember:**
> - Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
> - DeLancey: I suspension, II attachment (ATFP), III fusion.
> - MUS first line; TOMUS: retropubic = transobturator.
> - SISTEr: sling > Burch; PVS for ISD; urethral injury → no mesh.

Q3a. Sacral Neuromodulation.

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator).
- It acts on **afferents** to the forebrain, not motor nerves. In OAB (overactive bladder) it damps PMC (pontine micturition centre) firing; in retention it restores filling awareness, so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

Indications (third line, AUA (American Urological Association) and EAU (European Association of Urology))

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997	Bladder pain syndrome
Urgency–frequency, **non-obstructive retention** 1999	Neurogenic bladder (small studies positive)
Faecal incontinence	Pelvic pain

- **Fowler's syndrome** (young women in retention) responds well.

Technique

Test	PNE (percutaneous nerve evaluation)	Staged implant
Lead	Temporary, office, local anaesthesia	Permanent tined lead, theatre, fluoroscopy
Duration	5–7 days	Up to 4 weeks
Use	OAB	Higher conversion; retention

- Prone; S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot rotation
S3	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows only	None

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock; 10–16 Hz at sensory threshold.

Outcomes

- **InSite**: success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years.
- **ROSETTA**: onabotulinumtoxinA 200 U slightly better for UUI (−3.9 vs −3.3 episodes/day) and higher satisfaction, but more UTI (urinary tract infection) and self-catheterisation.

Complications and contraindications

- Pain at IPG site, lead migration, loss of efficacy, infection; **revision over 30% at 5 years**; costly; battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

Recent advances

- Rechargeable IPGs (≥15 years); **full-body MRI (magnetic resonance imaging)-conditional** systems; bilateral stimulation no better; PTNS (percutaneous tibial nerve

stimulation) as alternative.

Diagram to draw: lead in S3 foramen with IPG in the buttock.

> **Key points to remember:**
> - Afferent modulation treats OAB and retention.
> - S3: bellows plus great-toe flexion.
> - Implant if test improves ≥50%; revision >30% at 5 years.

Q3b. Botulinum toxin

Type: SAQ (10 marks)

Introduction

- *Clostridium botulinum* neurotoxin, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**; type B is short-acting.
- 150 kDa: heavy chain (100 kDa) plus zinc-dependent light chain (50 kDa).

Mechanism

1. Heavy chain binds the cholinergic terminal; toxin is internalised.
2. Light chain **cleaves SNAP-25 (synaptosomal-associated protein 25)**, a SNARE (vesicle-docking) protein, so vesicles cannot fuse and **acetylcholine release stops**.

- **Afferent action**: less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) receptors, so urgency falls.
- Effect wanes as terminals sprout (6–9 months).

Technique

- Cystoscopic injection into detrusor at **20 sites (100 U) or 30 sites (200 U)**, usually sparing trigone; sterile urine, antibiotic cover; PVR (post-void residual) at 2 weeks.

Urological uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK): incontinence −59%, about 25% dry. **ABC**: equal to antimuscarinic, more complete dryness. **ROSETTA**: 200 U ≥ SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY): lower detrusor pressure, larger capacity
Paediatric NDO (spina bifida)	About **10 U/kg (maximum 360 U)** in Campbell series; FDA label (2021, age ≥5) is 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation
DSD (detrusor–sphincter dyssynergia), underactive detrusor	Intrasphincteric 50–100 U; men unable to do CISC (clean intermittent self-catheterisation)
IC/BPS (interstitial cystitis/bladder pain syndrome)	100 U, often trigonal; AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	Intraprostatic: early small trial positive, but larger placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo**; not recommended

- **Also know:** paediatric dysfunctional voiding (sphincter); pelvic floor spasm; ED (erectile dysfunction) trials.

Complications

- **Retention needing CISC**: 0–12% in OAB trials; real-world about 20% women, up to 35% men; higher in NDO. Each 10 mL rise in baseline PVR raises risk about 27%. **Patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

- Contraindications: active UTI, myasthenia gravis, pregnancy. Repeat injections stay effective.

Diagram to draw: cholinergic synapse showing SNAP-25 cleavage.

> **Key points to remember:**
> - Cleaves SNAP-25; efferent and afferent action.
> - 100 U idiopathic OAB; 200 U NDO; children 10 U/kg (max 360 U) in series, FDA label 6 U/kg (max 200 U).
> - Main risks: retention needing CISC, UTI.

Q3c. Tacrolimus.

Type: SAQ (10 marks)

Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

Mechanism

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Tacrolimus + FKBP12 (FK506-binding protein 12)
   → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase)
   → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus
   → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)
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- Ciclosporin binds **cyclophilin** instead; same end result.

Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy)
Metabolic	**NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia
Neurological	**Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome)
Others	Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder))

Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

Versus ciclosporin

	Tacrolimus	Ciclosporin
Acute rejection	Lower	Higher
Diabetes, tremor, alopecia	More	Less
Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension	Less	More

- **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens.
- **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis).

Diagram to draw: T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - FKBP12 → calcineurin → NFAT → IL-2 blocked.
> - CYP3A4/P-gp; trough-guided; many interactions.
> - Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
> - ELITE-Symphony: low-dose tacrolimus best.

Q3d. Retrocaval ureter.

Type: SAQ (10 marks)

Definition

- The **right ureter** passes **behind the IVC (inferior vena cava)**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.

Embryology

- Paired fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney.
- Normally the right **supracardinal** vein, **dorsal** to the ureter, forms the infrarenal IVC, so the ureter lies in front.
- If a vein **ventral** to the ureter persists as the IVC (lumbar subcardinal in Campbell; "persistent posterior cardinal vein" in many texts), the ureter is trapped behind it.
- Variant: double right IVC (ureter between limbs).

Incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis. Associations: horseshoe kidney, left renal anomalies.

Classification (Bateson & Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower, at L3–L4	Higher, at renal pelvis level
Shape	**"Fishhook" / reverse-J**, S-curve	Smooth horizontal **sickle** curve
Obstruction	Moderate–severe, at iliopsoas edge	Mild or none (pelvis also crosses)

Investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) alternative.
- IVU (intravenous urography): fishhook, ureter lost beyond the J hook. Retrograde: **S-curve** at L3–L4. **Diuretic renogram**: obstruction, split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**; pyeloplasty fails.

Management

- **Non-obstructed (often type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it anterior to the IVC, spatulated tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; preserve blood supply.
- **Laparoscopic or robotic** (transperitoneal or retroperitoneal) now preferred: equal success, faster recovery. Nephrectomy if non-functioning.
- **Also know:** retroiliac ureter (L5–S1, vasal anomalies); mimics: spermatic or lumbar vein, psoas tendon.

Diagram to draw: right ureter hooking behind the IVC with fishhook pelvis.

Key points to remember:

- Venous anomaly; right side; 1 in 1500; male 3:1.
- Type I fishhook (obstructs); type II sickle.
- CT urography; repair by anterior transposition and ureteroureterostomy.

Q3e. DVIU (Direct Visual Internal Urethrotomy).

Type: SAQ (10 marks)

Definition and principle

- Incision of stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged.
- Healing is by **secondary intention**: a **race between epithelialisation and wound contraction**. If contraction wins, the stricture recurs. Deep spongiofibrosis cannot be cured endoscopically.

Indications (favourable)

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis, any non-bulbar
Short (<1.5–2 cm)	Longer
Superficial, no dense spongiofibrosis	Dense scar (straddle, LS (lichen sclerosus))
First procedure	Repeat; luminal narrowing >74%

Technique

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED (erectile dysfunction), fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit.

- **Laser** (holmium, Nd:YAG (neodymium:yttrium–aluminium–garnet), others): **no advantage over cold knife**.

Outcomes

- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none.
- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Repeat DVIU lengthens scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure.

Guideline position

- **AUA (American Urological Association) 2016:** DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **urethroplasty, not repeated endoscopy**, after failure; no permanent stents.
- Reconstructive ladder is abandoned.

Adjuncts after DVIU

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C, steroid injection (limited data); UroLume withdrawn.

Complications

- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI (urinary tract infection), sepsis; ED; spongio-cavernous fistula; incontinence if sphincter incised.

Recent advance

- **Optilume** paclitaxel-coated balloon (FDA (Food and Drug Administration) 2023): **ROBUST III** stricture-free 74.6% vs 26.8% at 1 year after failed endoscopy.

Diagram to draw: urethrotome cutting at 12 o'clock over guidewire.

> **Key points to remember:**
> - Best for bulbar, <1.5–2 cm, first attempt.
> - Santucci: ~8% long-term success; repeat DVIU futile.
> - AUA: urethroplasty after failure; laser = cold knife.

Q3f. TUR Syndrome.

Type: SAQ (10 marks)

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). About 1–2% in large series; falling.

Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, water (plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity**: ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Even routine resection absorbs 800–1000 mL.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (cooperative study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake patient: restlessness, confusion, **nausea, vomiting**, **visual disturbance**.
- Cardiovascular: **hypertension and bradycardia**, later hypotension, tachypnoea, ECG (electrocardiogram) changes.
- Severe: seizures, coma, death. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT (number needed to treat) about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time (stage large glands); check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol tagging** (breath test) monitors absorption.

Treatment

1. Stop surgery, achieve haemostasis, insert catheter.
2. ABC (airway, breathing, circulation); oxygen; monitor; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes up to 300 mL. **Target rise 4–6 mEq/L.**
5. Correction limit: up to **12 mEq/L in 24 h** for acute hyponatraemia (8–10 mEq/L if symptoms begin >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy; sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

Diagram to draw: flowchart: absorption → overload + hyponatraemia → cerebral oedema.

Key points to remember:

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

Rapid Revision Sheet – 2017-Winter Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe embryology of penis. Describe management of proximal hypospadias. (LAQ)

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias plus undescended testis → DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka preferred; fistula commonest complication.

Q2. Describe anatomy of pelvic floor in females and describe surgeries for SUI in females. (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- MUS first line; TOMUS: retropubic = transobturator.
- SISTEr: sling > Burch; PVS for ISD; urethral injury → no mesh.

Q3a. Sacral Neuromodulation. (SAQ)

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves ≥50%; revision >30% at 5 years.

Q3b. Botulinum toxin (SAQ)

- Cleaves SNAP-25; efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO; children 10 U/kg (max 360 U) in series, FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

Q3c. Tacrolimus. (SAQ)

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
- ELITE-Symphony: low-dose tacrolimus best.

Q3d. Retrocaval ureter. (SAQ)

- Venous anomaly; right side; 1 in 1500; male 3:1.
- Type I fishhook (obstructs); type II sickle.
- CT urography; repair by anterior transposition and ureteroureterostomy.

Q3e. DVIU (Direct Visual Internal Urethrotomy). (SAQ)

- Best for bulbar, <1.5–2 cm, first attempt.
- Santucci: ~8% long-term success; repeat DVIU futile.
- AUA: urethroplasty after failure; laser = cold knife.

Q3f. TUR Syndrome. (SAQ)

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.